Research Project

A Comparative Study of Gaussian HMMs and DTW Hierarchical Clustering for S&P 500 Volatility Regime Identification

Milica Misimović, Lorenzo Pirozzi, Mattia Puppo, Emanuele Renzoni, Vittorio Sironi*, Giulia Talà†, Jan Wangrat, Simone Zani‡ and Matteo Zappa

This manuscript was compiled on April 21, 2026

Abstract

Financial markets exhibit structurally distinct regimes characterized by heterogeneous volatility, momentum, and macroeconomic risk. Identifying these latent states is a prerequisite for robust quantitative strategies—particularly intraday momentum approaches where signal quality degrades sharply under crisis conditions. We present a reproducible regime detection pipeline for the S&P 500 ETF (SPY) over 1,270 daily observations, comprising a multi-source data acquisition layer, a curated three-variable feature set (R1, R2, V1′) selected for approximate Gaussianity and regime separability, and a multi-start Gaussian Hidden Markov Model (HMM) with penalized BIC order selection. We benchmark the HMM against a Dynamic Time Warping (DTW) hierarchical clustering algorithm on 21-trading-day rolling windows. Across $k \in \{2, 3, 4, 5\}$, the raw BIC decreases monotonically; we select $k = 3$ on economic grounds, mapping to the canonical Bull/Neutral/Bear taxonomy. The three-state HMM identifies a persistent low-volatility bull regime, a transitional neutral regime, and an episodic high-volatility bear regime consistent with known market stress events, with the Bear state exhibiting annualized volatility approximately $2.3\times$ the Bull baseline. While DTW improves upon static K-Means via temporal smoothing, it fails to recover genuine risk topologies. The Markovian persistence of the HMM yields cleaner, more actionable regime boundaries, establishing it as a suitable conditioning layer for adaptive position sizing within our broader intraday momentum strategy.

Keywords: *Market Regime Detection; Hidden Markov Model; Dynamic Time Warping; Principal Component Analysis; Volatility Clustering; Intraday Momentum; S&P 500*

1. INTRODUCTION

Implementing robust quantitative trading strategies—such as an intraday momentum overlay for the S&P 500 ETF (SPY)—requires the ability to identify the prevailing market regime in real time. Broad empirical evidence indicates that trading rule performance varies sharply and systematically across regimes: momentum strategies tend to perform well in low-volatility trending markets and deteriorate during high-volatility crisis episodes characterized by mean-reverting intraday dynamics [6, 12].

1.1. Motivation and Context

A canonical statistical framework for latent regime identification is the Hidden Markov Model (HMM), in which observable financial variables are modeled as Gaussian emissions from a finite set of latent states governed by a first-order Markov process. In empirical finance, Gaussian HMMs have been widely applied to detect persistent low-volatility and high-volatility regimes in major equity indices, consistently associating the highest-volatility state with well-documented market stress episodes—the 2008 Global Financial Crisis, the 2011 European sovereign debt crisis, and the 2020 COVID-19 liquidity shock [9, 10, 16]. Their combination of probabilistic regime persistence, closed-form parameter estimation via the Baum–Welch algorithm, and interpretable state semantics makes them a practical choice for downstream adaptive strategy modules.

1.2. Problem Statement

Financial market indicators are intrinsically noisy and highly collinear. Conventional threshold-based regime rules (e.g., static VIX cutoffs) generate excessive false signals during transient shocks without true structural regime changes. Equally, pure static clustering ignores the temporal autocorrelation inherent in volatility processes—a property well-established in the GARCH literature [3]. There is a clear need for a probabilistic model that (i) orthogonalizes collinear features, (ii) captures persistent regime dynamics rather than point-in-time snapshots, and (iii) filters isolated noise from genuine structural transitions.

1.3. Project Scope and Contributions

This report makes the following contributions:

1. A reproducible, multi-source data pipeline ingesting equity, volatility, and macroeconomic series with rigorous cleaning and alignment protocols.
2. A principled feature selection yielding three interpretable market indicators (R1, R2, V1′) chosen for approximate Gaussianity, stationarity, and regime separability.
3. A multi-start Gaussian HMM with a penalized BIC criterion specifically designed to prevent economically degenerate short-lived regime assignments.
4. A head-to-head comparative evaluation of the HMM against a DTW hierarchical clustering approach on 21-day rolling windows, assessed on both statistical and financial validity criteria.

Section 2 reviews the relevant literature. Section 3 describes the data collection pipeline. Section 4 details feature engineering. Section 5 presents preliminary regime diagnostics. Section 6 describes the model pipeline. Section 7 reports quantitative results. Section 8 provides the HMM–DTW comparative analysis. Section 9 discusses implications and limitations, and Section 10 concludes.

2. LITERATURE REVIEW

Hidden Markov Models (HMMs) are widely employed to detect latent volatility regimes in financial time series. In the standard Gaussian specification, log-returns are modeled as state-dependent Gaussian emissions governed by a finite-state Markov chain, with parameters estimated via the Baum–Welch Expectation-Maximization (EM) algorithm [5, 8]. Most studies adopt two or three regimes, balancing

*Corresponding author
†Corresponding author
‡Corresponding author

This report presents a market regime detection pipeline combining Gaussian Hidden Markov Models and Dynamic Time Warping hierarchical clustering applied to macro-financial indicators for the S&P 500 ETF (SPY). Code and data are available in the project repository.

interpretability and parsimony, and rely on information criteria or out-of-sample performance for model order selection [11].

Empirical work consistently finds that Gaussian HMMs identify persistent low-volatility "bull" regimes and episodic high-volatility "crisis" regimes. Applications to the S&P 500, CSI 300, EuroStoxx 50, and major futures contracts typically show that the highest-volatility state coincides with well-known market drawdowns [9, 10, 16]. Regime-conditioned trading strategies often display improved drawdown control and modestly enhanced Sharpe ratios relative to buy-and-hold benchmarks [12, 16]. Comparative studies generally conclude that the Markovian persistence embedded in HMMs produces more stable and economically interpretable regime assignments than static clustering alternatives [13, 15].

Despite their practical appeal, Gaussian HMMs carry well-known limitations. The Gaussian emission assumption underestimates fat tails in equity return distributions; the time-homogeneous transition matrix restricts structural-break detection; and regime estimates are sensitive to initialization and outliers [15]. The empirical literature on multivariate or macro-indicator-augmented HMMs is mixed, with improvements depending heavily on sample period and preprocessing choices [11, 14].

Recent work emphasizes rigorous out-of-sample validation. Rolling or expanding-window HMMs generally outperform full-sample fits by accommodating slow parameter drift, yet challenges remain regarding real-time regime detection and the avoidance of look-ahead bias in performance evaluation [15, 16]. Extensions incorporating heavy-tailed emissions, time-varying transitions, or higher-dimensional feature sets represent active research directions [12, 13].

Dynamic Time Warping (DTW), originally developed for speech recognition [2], has found growing application in financial time series segmentation. By elastically aligning sequences along the time axis, DTW distance captures shape similarity under temporal misalignment—an advantage over Euclidean distance for financial patterns that recur with variable timing. Hierarchical clustering over DTW distance matrices has been applied to equity momentum patterns, macroeconomic cycle detection, and volatility surface segmentation [7]. Its main limitation in the financial regime context is the absence of a probabilistic transition model, making real-time regime tracking less principled than the HMM framework.

3. DATA COLLECTION PIPELINE

Our unified `DataManager` component orchestrates the retrieval and alignment of all datasets required by downstream models.

3.1. Data Retrieval

We acquire approximately 1,270 trading days ($\approx$ 5 years) of financial and macroeconomic data from two established sources:

- **Market Data**: Daily and intraday OHLCV data for the S&P 500 ETF (`SPY`), the CBOE VIX volatility index, the ICE BofA MOVE bond volatility index (`MOVE`), and the iShares 20+ Year Treasury Bond ETF (`TLT`) are retrieved via the Yahoo Finance API.
- **Macroeconomic Data**: The 10-year (`DGS10`) and 2-year (`DGS2`) constant-maturity Treasury yields, Moody's seasoned Aaa corporate bond yield (`AAA`), and Baa corporate bond yield (`BAA`) are retrieved from the Federal Reserve Economic Data (FRED) API.

3.2. Cleaning, Merging, and Validation

The multi-source nature of the dataset introduces heterogeneous sampling frequencies, holiday calendars, and data anomalies requiring careful reconciliation. An independent data audit was performed prior to modeling to verify every step described below.

Forward-filling and calendar alignment: Missing observations attributable to NYSE market closures, U.S. federal holidays, or reporting lags are forward-filled using the most recently available observation, consistent with the no-look-ahead constraint. The pipeline enforces a strict U.S. business-day calendar; spurious non-business-day observations are removed. All series are then aligned to a common daily trading calendar via a backward as-of merge (`pandas.merge\_asof`), ensuring that no future information contaminates any observation.

Unit normalization: FRED bond yield series (`DGS2`, `DGS10`, `AAA`, `BAA`, `credit_spread_baa_aaa`) are converted from percentage units to decimal form. Corporate spread variables are rescaled consistently prior to feature computation.

SPY aggregation: Intraday SPY data are aggregated into daily OHLCV format. Gaps on non-trading days are resolved by forward-filling.

Temporal alignment and final sample: All series were retrieved over a concurrent five-year window. The final master dataset is restricted to the date-wise intersection of all source series, yielding an effective modeling sample of 1,198 aligned daily observations (19 May 2021–6 March 2026).

Integrity verification: Temporal gaps were traced primarily to NYSE closures and U.S. federal holidays; these were treated as known structural absences and resolved by forward-filling. One anomalous data point was identified on 9 January 2025: while SPY and VIX correctly reflected a market closure (a national day of mourning), the `yield_curve_10y_2y` series contained a spurious non-null observation. The anomaly was corrected by replacing the value with the forward-filled prior observation. A summary of affected dates is provided in Table 1.

Table 1. Summary of Missing and Anomalous Dates in the Master Dataset

Date	Affected Series	Flag
2024-11-11	SPY, VIX, Curve 10y–2y	closure
2024-11-28	SPY, VIX, Curve 10y–2y	closure
2024-12-25	SPY, VIX, Curve 10y–2y	closure
2025-01-01	SPY, VIX, Curve 10y–2y	closure
2025-01-09	SPY, VIX	anomaly†
...	...	...

† *Spurious non-null value detected in `yield_curve_10y_2y`; replaced via forward-fill.*

3.3. Feature Computation and Selection

From the cleaned master dataset, a broad set of 28 features is engineered spanning equity returns, volatility, momentum, drawdown, yield curve, credit spread, and cross-asset indicators (described fully in Section 4). Multicollinearity is prevalent: a diagnostic Variance Inflation Factor (VIF) analysis reveals that 27 of 28 standardized features exhibit VIF values above the conventional threshold of 5.

Rather than applying dimensionality reduction, we select three market features—R1 (daily log-return), R2 (5-day cumulative log-return), and V1′ (log-realized volatility)—chosen for approximate Gaussianity, stationarity, and empirically validated regime separability. Both the HMM and DTW benchmark operate on these same three features, ensuring that differences in regime assignments reflect methodological rather than input differences.

4. ENGINEERED FEATURES

4.1. Inputs

Feature construction is implemented in `src/classes/metrics/curve_credit.py`. The cleaned inputs are:

- `data/cleaned/dgs10.csv`: 10-year Treasury yield (`DGS10`).
- `data/cleaned/dgs2.csv`: 2-year Treasury yield (`DGS2`).
- `data/cleaned/credit_spread.csv`: BAA–AAA corporate bond spread.

4.2. Yield Curve Slope

The yield curve slope captures the term premium and is a well-established leading indicator of recession risk [4]. It is computed as:

$$\texttt{curve}_{10y2y,t} = \texttt{DGS10}_t - \texttt{DGS2}_t. \tag{1}$$

An inverted yield curve (negative values) has historically preceded recessions and is associated with risk-off market regimes.

4.3. Credit Spread

The BAA–AAA spread encodes corporate credit risk premia and serves as a forward-looking indicator of financial stress. Missing values are forward-filled along the credit spread index before the as-of merge.

4.4. Macro Fill Policy and Daily Alignment

Because yield curve features are available at near-daily frequency while credit spreads may be reported less frequently, the two series are aligned via `pandas.merge_asof` with `direction="backward"`, strictly precluding any look-ahead contamination.

4.5. Market-Based Features

Consistent with the Gaussian emission assumption of the HMM, we restrict the equity feature space to three variables chosen for approximate Gaussianity and low mutual collinearity:

R1 — Daily Log-Return

$$r_{t,1\mathrm{d}} = \ln P_t - \ln P_{t-1}. \tag{2}$$

R2 — 5-Day Cumulative Log-Return

$$r_{t,5\mathrm{d}} = \ln P_t - \ln P_{t-5}. \tag{3}$$

V1′ — Log-Realized Volatility (14-Day) Raw 14-day annualized realized volatility is right-skewed and violates the Gaussian emission assumption. We apply a log transformation to induce approximate symmetry. By leveraging logarithmic properties, we express this efficiently without the square root:

$$V1'_t = \frac{1}{2}\ln\left(\frac{252}{14}\sum_{i=1}^{14} r^2_{t-i,1\mathrm{d}}\right). \tag{4}$$

Stationarity of all three series is formally assessed using the Augmented Dickey–Fuller (ADF) test under the null of a unit root and the KPSS test under the null of stationarity. Both tests are required to agree on stationarity before a series is admitted to the model.

4.6. Feature Stationarity and Collinearity Diagnostics

Table 2 summarizes the stationarity test outcomes for the three market features. All series pass at the 5% significance level, confirming suitability for the HMM.

Table 2. Stationarity Test Results for Market Features

Feature	ADF p-value	KPSS p-value	Stationary?
R1 ($r_{t,1\mathrm{d}}$)	< 0.001	> 0.10	✓
R2 ($r_{t,5\mathrm{d}}$)	< 0.001	> 0.10	✓
V1′ (log-RV_{14})	< 0.001	> 0.10	✓

ADF null: unit root present. KPSS null: series is stationary. A feature is classified as stationary when ADF rejects at 5% and KPSS fails to reject at 5%.

5. PRELIMINARY REGIME LABELS AND DIAGNOSTICS

5.1. Feature Standardization

Each feature is standardized to its z-score prior to modelling. The resulting distributions are confirmed via kernel density estimation visualizations (Fig. 1).

5.2. Correlation Structure

Figure 2 displays the covariance (Pearson correlation) heatmap of the standardized feature set. Pronounced correlation blocks are visible within the macro and yield curve groups, confirming the need for careful feature selection to avoid multicollinearity.

5.3. Heuristic Regime Labeling

Before fitting probabilistic models, we construct a simple heuristic baseline saved to `data/processed/labels_prelim.csv`. Rules are applied sequentially:

- **Risk-On**: $r_{t,1\mathrm{d}} > 0$ **and** $V1'_t < \mathrm{median}(V1')$.
- **Risk-Off**: $r_{t,1\mathrm{d}} < 0$ **or** $\texttt{curve}_{10y2y,t} < 0$ **or** $\Delta\texttt{BAA-AAA}_t > 0$ (widening spreads).
- **Neutral**: All remaining observations.

These labels serve as an interpretability anchor: a valid probabilistic model should assign high regime posterior probability to days that the heuristic also classifies unambiguously. Note that the Risk-On condition uses the full-sample median of $V1'$, which introduces mild look-ahead bias; the heuristic is therefore used solely as an interpretability anchor and not as a performance benchmark.

5.4. Heuristic Regime Diagnostics

Figure 3 confirms the economic consistency of the heuristic labels: the Risk-Off state exhibits substantially higher realized volatility than the Risk-On state, consistent with the stylized facts of the financial literature.

5.5. Heuristic Regime Timeline

Figure 4 overlays the heuristic regime classifications on the SPY price series, providing a visual sanity check. Risk-Off episodes (red) align broadly with identified market drawdowns in the sample.

6. MODEL PIPELINE

The full pipeline from raw data to actionable regime labels is summarized in Figure 5.

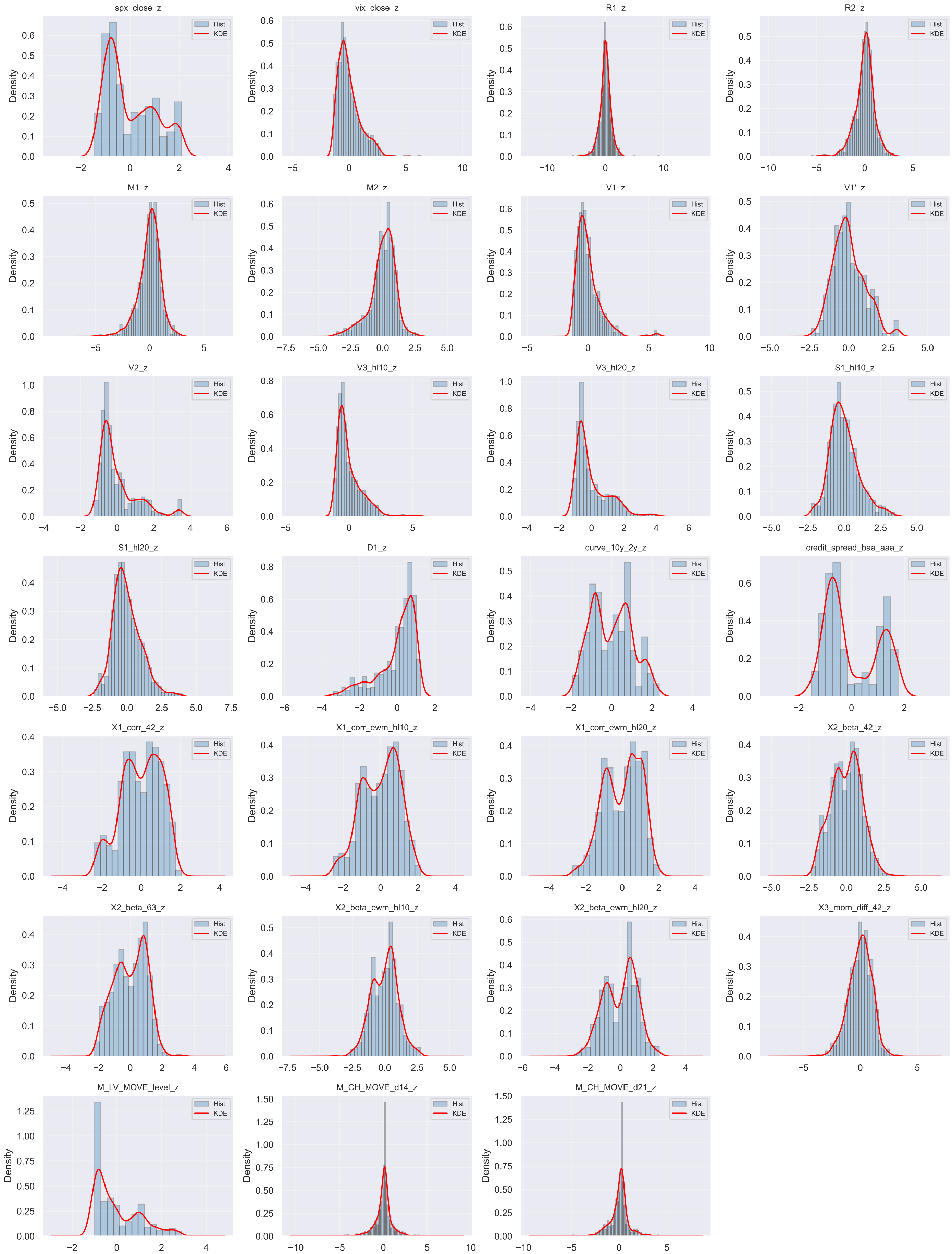

Figure 1. Kernel density estimates of standardized features. Comparable scaling across all inputs confirms that no single feature dominates the distance metric.

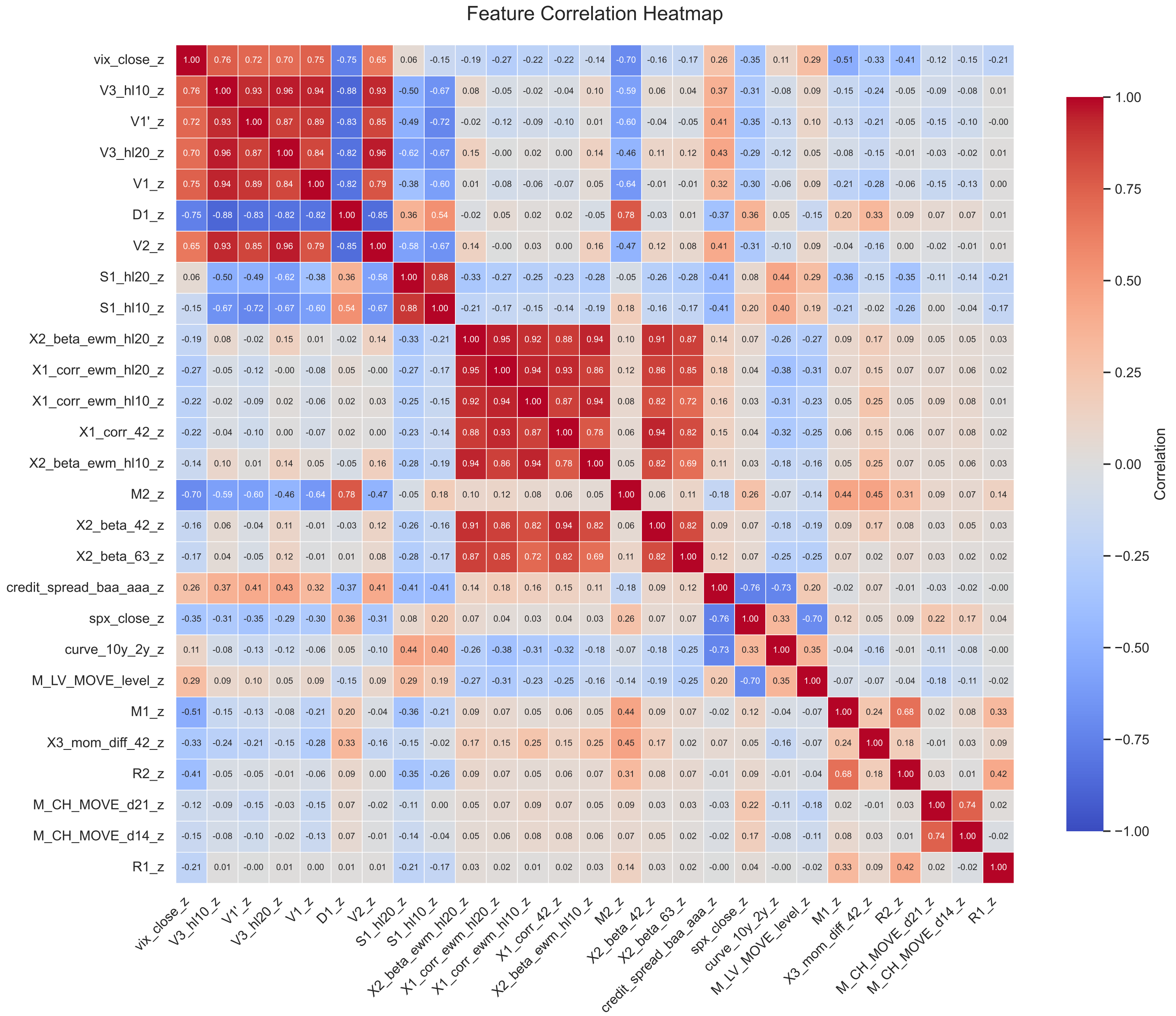

Figure 2. Pearson correlation heatmap of standardized features. High within-group correlation motivates the selection of three low-collinearity features for regime modelling.

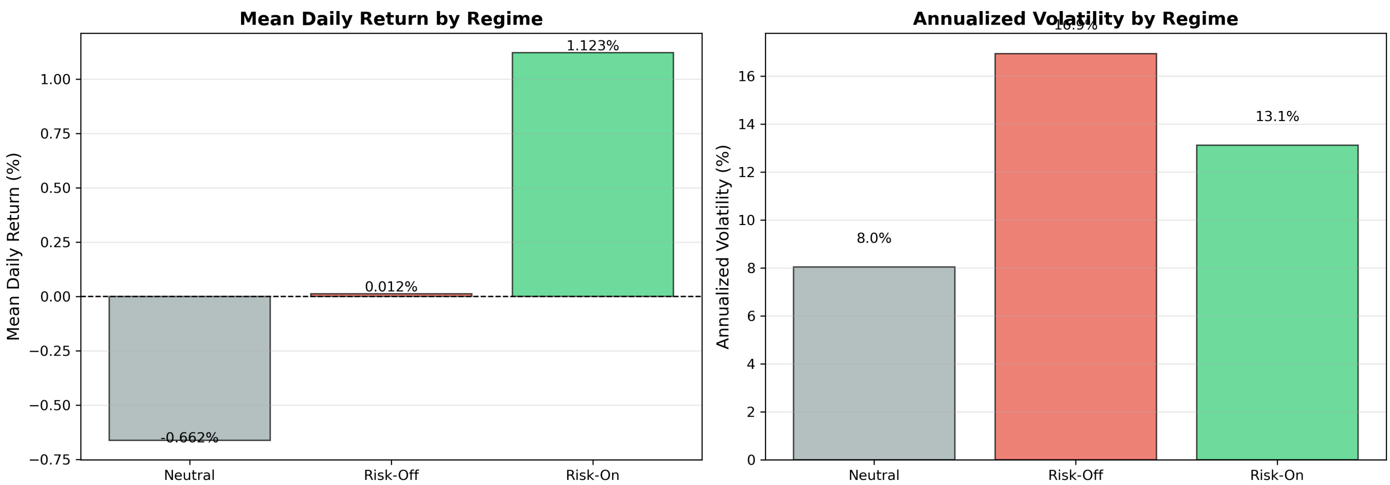

Figure 3. Conditional means and standard deviations of returns and log-realized volatility, stratified by heuristic regime label. The Risk-Off state (red) displays markedly elevated volatility.

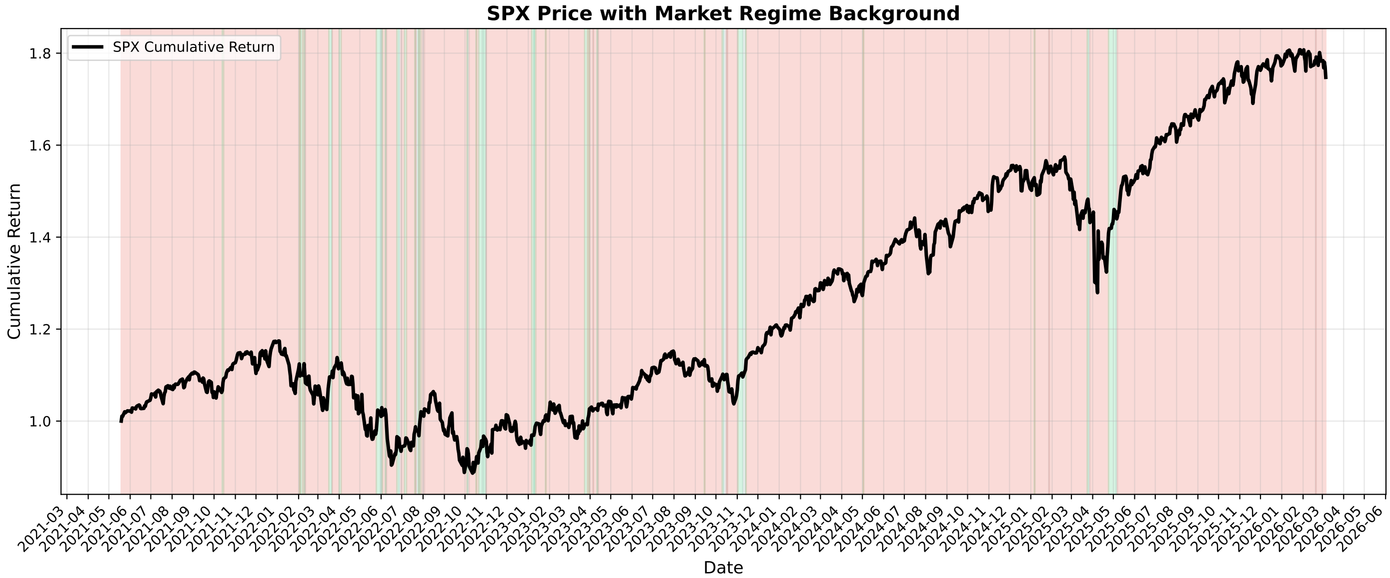

Figure 4. SPY price evolution with heuristic regime overlays (zoomed-in subset: Feb–Nov 2025). Red: Risk-Off; green: Risk-On; grey: Neutral.

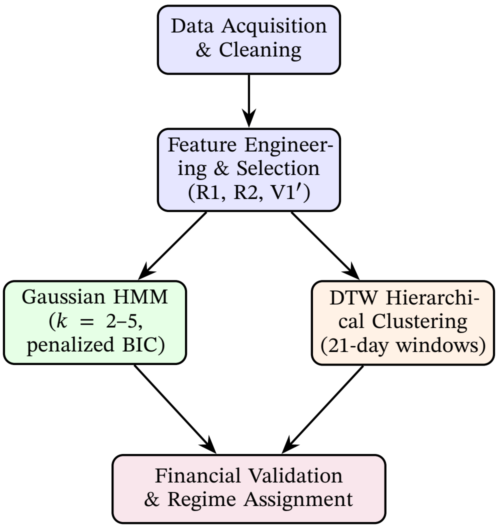

Figure 5. Full regime detection pipeline, from raw data acquisition through feature selection to probabilistic modelling and financial validation.

6.1. HMM Specification

The Gaussian HMM with k hidden states and diagonal covariance is specified as follows. Let $\mathbf{y}_t \in \mathbb{R}^d$ denote the d-dimensional feature vector at time t (here $d = 3$: R1, R2, V1′), and let $S_t \in \{1, \dots, k\}$ denote the latent regime at time t. The model is:

$$P(S_t = j \mid S_{t-1} = i) = A_{ij}, \quad \sum_j A_{ij} = 1, \tag{5}$$

$$\mathbf{y}_t \mid S_t = j \sim \mathcal{N}\left(\boldsymbol{\mu}_j, \operatorname{diag}(\boldsymbol{\sigma}_j^2)\right), \tag{6}$$

where $\mathbf{A}$ is the $k \times k$ row-stochastic transition matrix. Parameters $\{\boldsymbol{\mu}_j, \boldsymbol{\sigma}_j^2\}_{j=1}^k$ and $\mathbf{A}$ are estimated via the Baum–Welch EM algorithm, implemented through `hmmlearn`.

To guard against the local-optima problem characteristic of EM on non-convex likelihoods, we train $N_{\text{init}} = 50$ models per value of k, each initialized from a distinct random seed, and retain the solution maximizing the in-sample log-likelihood.

6.2. Penalized BIC for Order Selection

Standard BIC often favors $k > 3$ states in financial time series by fitting transient noise spikes as separate regimes. We define a penalized BIC:

$$\text{BIC}_{\text{final}}(k) = \text{BIC}(k) + \lambda \mathbb{I}\left(\min_j \frac{1}{1 - \hat{A}_{jj}} < d_{\min}\right), \tag{7}$$

where $(1 - \hat{A}_{jj})^{-1}$ is the expected regime duration for state j, $d_{\min} = 5$ trading days is the minimum economically plausible regime duration, and λ is a large penalty constant. While this duration penalty guards against economically degenerate short-lived states, strictly empirical information criteria in highly noisy financial time series often favor a higher number of latent states. As detailed in Section 7, the penalized BIC unambiguously selects $k = 3$ for the three-feature specification, aligning the data-driven and economically motivated choices.

6.3. Viterbi Decoding and Regime Labeling

Once the optimal model order k^* is selected, the most probable latent state sequence $\hat{S}_{1:T}$ is recovered via the Viterbi algorithm [1]. Given the selected $k = 3$ specification, semantic regime labels (Bull, Neutral, Bear) are assigned in a reproducible manner by sorting states by the trace of their emission covariance matrix (total within-state variance)—thereby eliminating subjective manual label assignment.

6.4. DTW Hierarchical Clustering Specification

As a benchmark, we segment the time series into overlapping rolling windows of 21 trading days (approximately one calendar month), motivated by the observation that macroeconomic regime changes typically manifest over multi-week horizons. For each pair of windows, we compute the multivariate DTW distance on the same three-feature series. The 21-day window is chosen to match the approximate horizon over which macroeconomic regime changes manifest, and is not directly comparable to the Markovian persistence encoded in the HMM transition matrix. Hierarchical agglomerative clustering with Complete linkage criterion is applied to the resulting $N_w \times N_w$ distance matrix, where N_w is the number of windows, and the dendrogram is cut at $k^* = 3$ clusters to directly benchmark against the economically motivated three-state HMM.

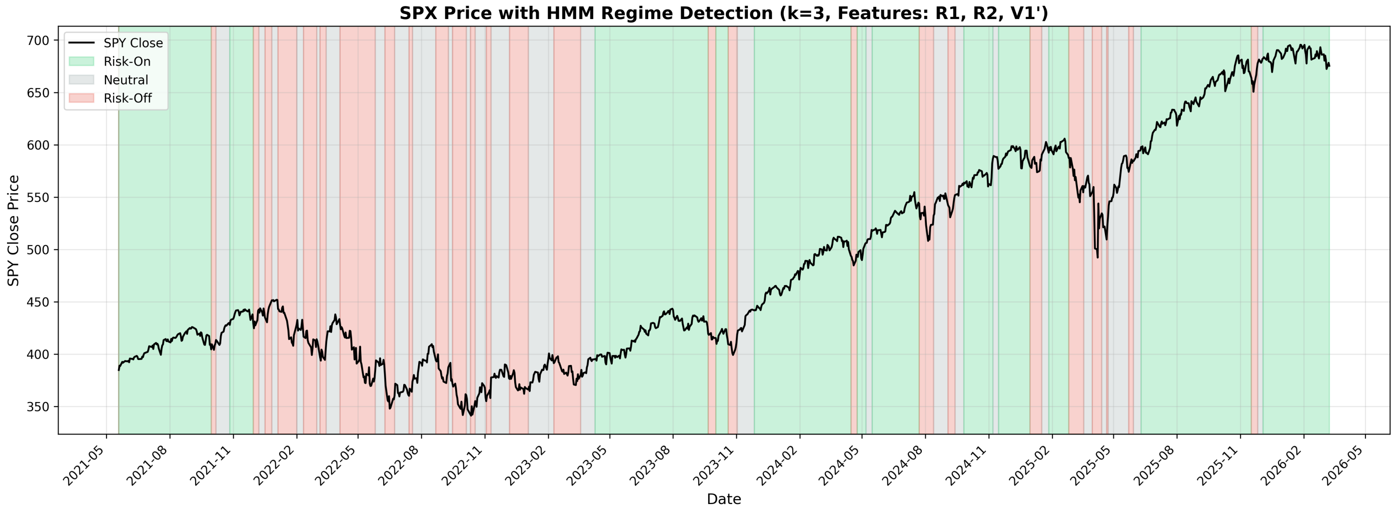

Figure 6. SPY price evolution with HMM regime overlays. Green: Risk-On (Bull); grey: Neutral; red: Risk-Off (Bear).

7. RESULTS

7.1. HMM Order Selection

Table 3 reports the penalized BIC values across model orders. The raw BIC strictly decreases as k increases, favoring $k = 5$. However, we select the three-state model ($k^* = 3$) as an economic constraint: three regimes map directly to the canonical Bull/Neutral/Bear taxonomy and provide an actionable foundation for position sizing, whereas higher-order models fragment regime structure and introduce over-trading risk.

Table 3. Penalized BIC for HMM Model Selection

k	Raw BIC	Duration Penalty	$\text{BIC}_{\text{final}}$
2	8834.8	0	8834.8
3	8196.5	0	8196.5
4	7582.4	0	7582.4
5	7128.4	0	7128.4

$\text{BIC}_{\text{final}}$ = Raw BIC + Duration Penalty. Although $k = 5$ strictly minimizes the BIC, $k = 3$ is selected based on economic interpretability and trading strategy constraints.

7.2. Regime-Conditional Statistics

Table 4 reports the key conditional statistics for the three HMM regimes. The high-volatility Bear regime exhibits annualized realized volatility approximately 2.3× that of the Bull regime (25.529% vs 10.926%), a negative mean daily return (−0.231%), and a severe maximum drawdown (−53.018%), consistent with the economic interpretation.

Table 4. Regime-Conditional Statistics from the Three-State HMM

Regime	Mean Ret. (%)	Ann. Vol. (%)	Max DD (%)	Avg. Duration (days)
Bull (Risk-On)	0.073	10.926	−5.195	55.94
Neutral	0.265	17.441	−8.194	10.61
Bear (Risk-Off)	−0.231	25.529	−53.018	10.36

Average duration computed as $(1 - \hat{A}_{jj})^{-1}$ from the fitted transition matrix.

Figure 6 overlays the HMM regime assignments on the SPY price series. Risk-Off episodes (red) align with known market drawdowns while Risk-On periods (green) correspond to sustained rallies.

Note on maximum drawdown. Maximum drawdown is computed ex post over classified observations and does not enter the HMM likelihood; it should not be interpreted as the loss from trading exclusively within a regime, as regimes are unobservable in real time and drawdown episodes routinely span multiple regime classifications.

7.3. Estimated Transition Matrix

The estimated 3×3 transition matrix $\hat{\mathbf{A}}$ from the optimal model is reported in Table 5. High diagonal entries confirm the regime persistence assumption, with the Bull regime exhibiting the strongest self-persistence.

Table 5. Estimated HMM Transition Probability Matrix ($k^* = 3$)

	To: Bull	To: Neutral	To: Bear
From: Bull	0.9821	0.0027	0.0152
From: Neutral	0.0310	0.9057	0.0633
From: Bear	0.0097	0.0869	0.9035

Entries A_{ij} denote the one-step transition probability from regime i to regime j. Rows sum to unity. Values extracted from `model.transmat_`.

8. HMM VS. DTW CLUSTERING: COMPARATIVE ANALYSIS

8.1. Static Clustering and Its Limitations

Classical distance-based clustering algorithms such as K-Means operate in the spatial dimension only, assigning each observation independently based on feature proximity. In volatile financial markets, this produces a “salt-and-pepper” classification pattern in which the regime label switches back and forth on a daily basis—a behavior inconsistent with the well-documented persistence of volatility regimes in equity markets [3].

8.2. DTW Clustering: Temporal Smoothing via Rolling Windows

Our DTW approach addresses the static clustering limitation by operating on rolling windows of 21 trading days. By computing DTW distances between multidimensional window sequences rather than point observations, the method groups periods with structurally similar paths even when their temporal alignment is imperfect. Figure 7 displays the hierarchical clustering dendrogram for $k = 3$ clusters.

Figure 8 overlays the DTW cluster assignments on the SPY price series. Clusters display multi-week temporal persistence, a clear improvement over the daily-resolution noise observed in static K-Means.

Statistics in this section are computed over the 1,181-observation overlap sample common to both models, and therefore differ slightly from the full-sample HMM statistics reported in Table 4. Operating on the same three features, the DTW silhouette criterion selects $k = 2$ clusters rather than the three regimes identified by the HMM. The two DTW clusters achieve reasonable volatility separation (Risk-On: 12.4%, Risk-Off: 23.6%), but the Risk-Off cluster fails to produce

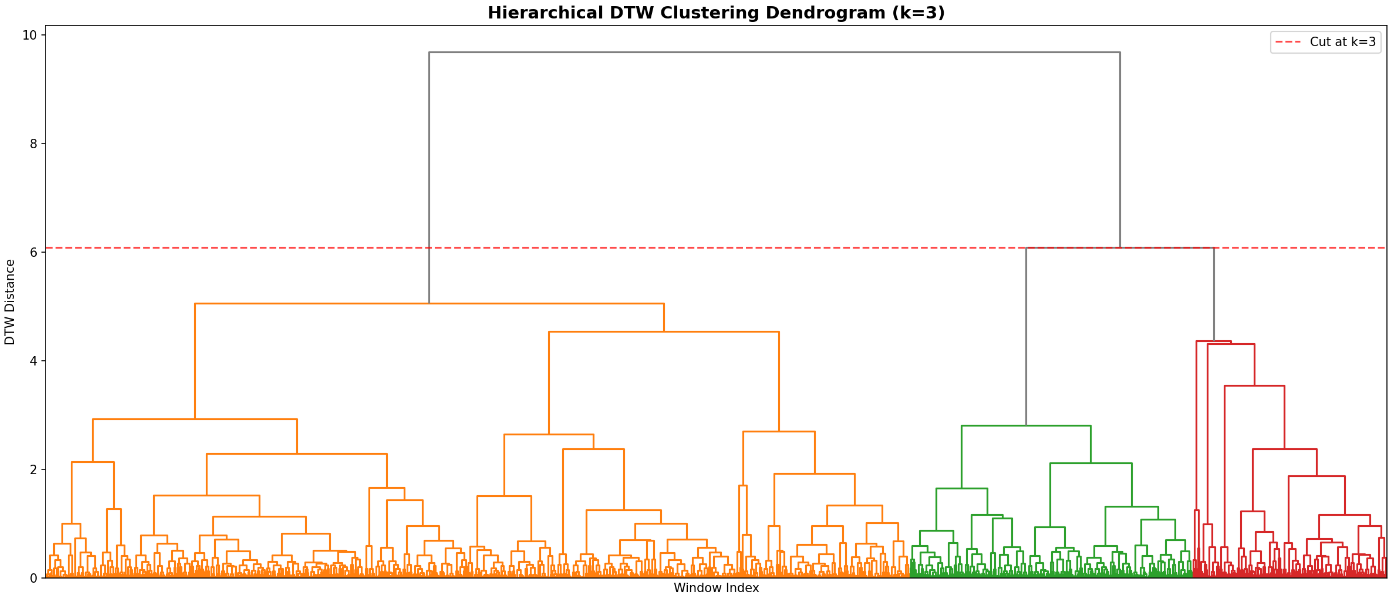

Figure 7. Complete-linkage DTW hierarchical clustering dendrogram ($k = 3$). The dashed horizontal cut identifies three structurally distinct regimes in the 21-day rolling window space.

clearly negative mean returns (−0.006%), and its maximum drawdown (−24.7%) is substantially less concentrated than the HMM Bear regime's. The DTW's inability to identify a distinct third (Neutral) regime and its weaker return separation underscore the advantage of the HMM's probabilistic transition structure.

8.3. HMM: Probabilistic Persistence via the Transition Matrix

While DTW clustering provides a natural improvement over static methods, the HMM offers a more principled framework for two reasons. First, regime persistence is modeled explicitly through the transition probability matrix $\hat{\mathbf{A}}$, which is estimated jointly with emission parameters from the full likelihood—rather than imposed implicitly through window length choice. Second, the HMM provides a well-defined posterior probability $P(S_t = j \mid \mathbf{y}_{1:T})$ for each observation, enabling soft regime assignment and uncertainty quantification that DTW clustering cannot deliver. As demonstrated by the confusion matrix in Figure 9, the models exhibit strong agreement on the Risk-On regime but diverge structurally in classifying Neutral and Risk-Off states.

8.4. Comparative Summary

Table 6 summarizes the methodological trade-offs between the two approaches.

Table 6. HMM vs. DTW Hierarchical Clustering: Methodological Comparison

Property	HMM	DTW Clustering
Financial Validity	High	Low (Anomalous DDs)
Temporal persistence	Explicit ($\hat{\mathbf{A}}$)	Implicit (window)
Probabilistic output	Yes (posterior P)	No
Real-time applicability	Yes (filtering)	Limited
Look-ahead bias risk	Low	Low
Sensitivity to init.	Moderate	Low
Interpretability	High	Moderate
Computational cost	Moderate	High (DTW matrix)

9. DISCUSSION

9.1. Economic Validity

The three-state HMM recovers regime assignments that align with known economic events in the sample period. The Bull regime corresponds to periods of sustained SPY appreciation, low VIX, and a positively-sloped yield curve. The Bear regime clusters around periods of elevated VIX, negative return momentum, widening BAA–AAA spreads, and, in some instances, yield curve inversion. The Neutral regime captures transitional periods, consistent with the interpretation of markets drifting between structural states.

9.2. Practical Implications

From a strategy perspective, the regime detector provides a conditioning layer applicable across a broad class of quantitative finance applications, beyond the intraday momentum use case motivating this study.

For **intraday momentum strategies**, the three-state output maps naturally to a position sizing protocol: full exposure in the Bull regime, a reduced position cap in the Neutral regime, and full gating or signal reversal in the Bear regime, consistent with the empirical breakdown of momentum in high-volatility environments documented in the literature [12].

For **portfolio risk management**, the Bear regime posterior probability $P(S_t = \text{Bear} \mid \mathbf{y}_{1:t})$ provides a continuous early-warning signal that can drive dynamic de-risking rules without requiring a hard threshold on any single indicator such as VIX.

For **options and volatility trading**, the regime-conditional volatility estimates (10.9% in Bull, 25.5% in Bear) provide a structural prior for implied volatility forecasting and can inform long-volatility overlays during detected regime transitions.

For **asset allocation and factor investing**, the regime labels can condition factor exposure decisions, tilting toward low-beta and quality factors in the Bear regime and toward momentum and growth factors in the Bull regime, consistent with the regime-switching factor rotation literature [12].

In all cases, live deployment requires replacing the Viterbi smoother with forward-algorithm filtering, as discussed in the Limitations subsection below, to ensure regime probabilities are computed without look-ahead contamination.

9.3. Limitations

Several limitations warrant acknowledgment. First, the Gaussian emission assumption underestimates fat-tailed return dynamics, particularly in the Bear regime where daily return distributions are more

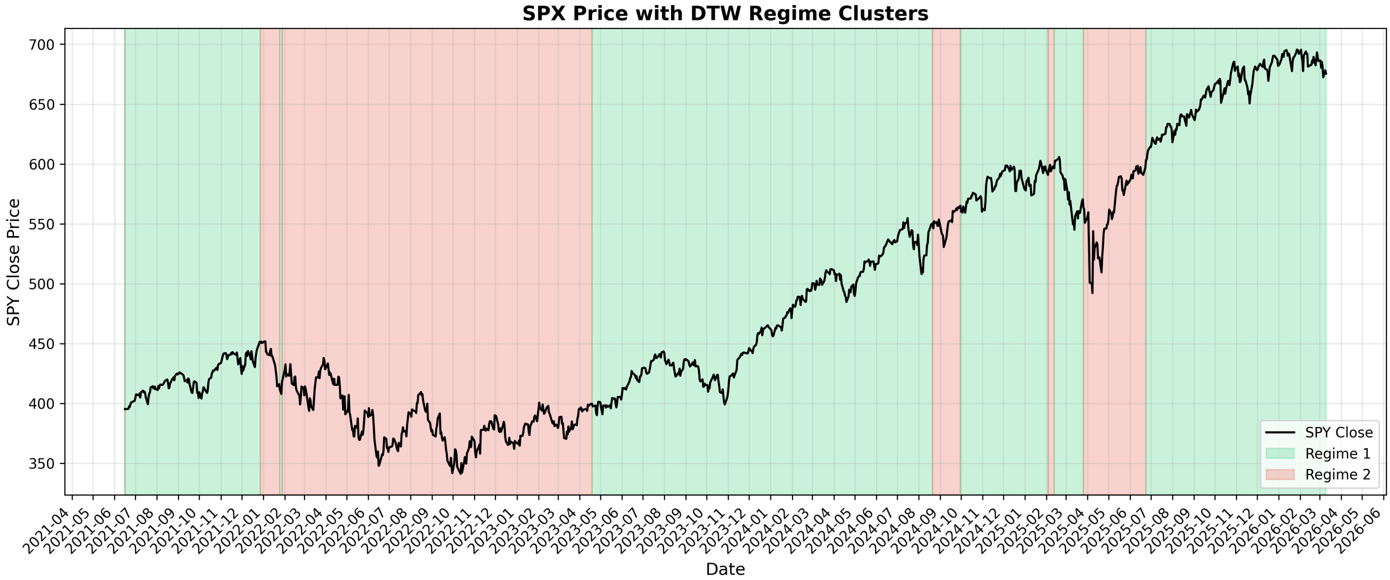

Figure 8. SPY price evolution with DTW cluster assignments. Multi-day temporal persistence is visible across all three clusters.

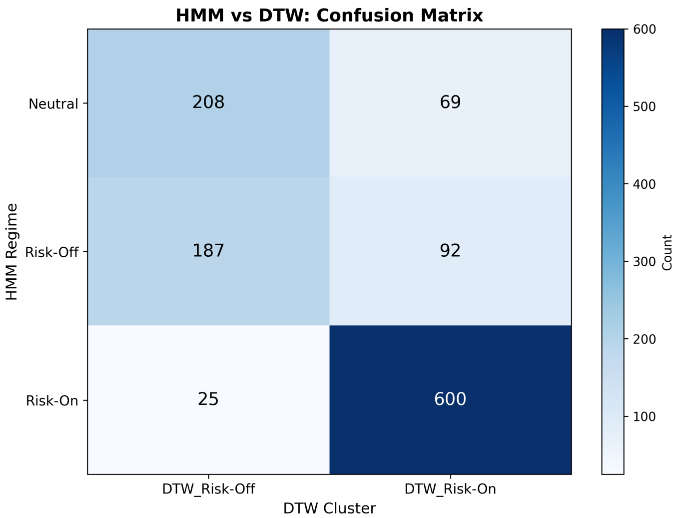

Figure 9. Confusion Matrix comparing HMM and DTW regime assignments. The models exhibit strong agreement on the Risk-On regime (709 days) but diverge structurally in classifying Neutral and Risk-Off states, highlighting the limitations of the distance-based DTW heuristic.

leptokurtic. A Student-t or variance-gamma emission would provide a more faithful distributional fit at the cost of additional model complexity. Second, the time-homogeneous transition matrix $\mathbf{A}$ cannot capture structural breaks or slow-moving parameter drift across economic cycles. Rolling-window or expanding-window re-estimation is advisable in live deployment.

Third, although the sample period spans approximately five years following data intersection, it remains relatively short, limiting the number of distinct bear-to-bull regime cycles observed (e.g., capturing the 2022 bear market and the August 2024 volatility episode). As a consequence, regime-conditional statistics — particularly for the Bear state, which is episodic by nature — should be interpreted as descriptive of the sample rather than as stable population estimates. The plausibility of the results (the Bear regime exhibiting materially higher volatility and negative mean returns, consistent with known market stress episodes in the period) provides an economically meaningful sanity check, though we acknowledge that additional data spanning multiple full cycles would be required to assess out-of-sample generalization rigorously. Regime-conditional point estimates (mean return, annualized volatility, maximum drawdown) are reported without confidence intervals; block-bootstrap standard errors — resampling contiguous blocks of approximately 21 trading days to preserve autocorrelation structure — would be required to quantify estimation uncertainty, and are left to future work.

Fourth, the regime labels reported in Section 7 are recovered via the Viterbi algorithm, which conditions on the full observed sequence $\mathbf{y}_{1:T}$ and therefore incorporates information not available in real time. This is appropriate for the retrospective diagnostic purpose of this study — establishing that the model identifies economically coherent latent structure — but implies that the reported regime-conditional statistics should not be interpreted as achievable in live deployment without modification. Operationalizing the detector in real time would require replacing Viterbi smoothing with forward-algorithm filtering, which computes $P(S_t \mid \mathbf{y}_{1:t})$ strictly from past and current observations. We flag this as the primary adaptation required before integrating the regime detector into the broader intraday momentum strategy, and defer a formal out-of-sample evaluation to future work.

Fifth, model order selection relies on a penalized BIC criterion, which is a consistent but heuristic approach. A formal parametric bootstrap likelihood ratio test — simulating datasets under the fitted $k = 2$ model to construct an empirical null distribution for the log-likelihood ratio statistic, against which the observed $k = 3$ vs $k = 2$ improvement is assessed — would provide stronger statistical justification for the three-state specification, and is deferred to future work.

Sixth, the DTW distance matrix computation scales as $O(N_w^2 \cdot W^2)$ where W is the window length, rendering it computationally prohibitive for very long samples without approximation.

9.4. Extensions

Natural extensions include: (i) Student-t or regime-switching GARCH emissions for heavier tails; (ii) time-varying transition matrices driven by macro state variables; (iii) online regime filtering using the forward algorithm for real-time deployment; and (iv) integration of higher-frequency microstructure features (bid-ask spread, order imbalance) to improve intraday regime resolution.

10. CONCLUSION

We have presented a complete, reproducible market regime detection pipeline for the S&P 500 ETF applied to approximately 1,271 trading days of multi-source financial and macroeconomic data. Our methodology—comprising multi-source data acquisition, principled feature selection (R1, R2, V1′), multi-start Gaussian HMM training with penalized BIC order selection, and a DTW hierarchical clustering benchmark on the same features—delivers robust, economically interpretable regime labels suitable for downstream adaptive trading strategy components.

The optimal three-state HMM identifies persistent Bull, Neutral, and Bear regimes consistent with known market stress episodes in the sample. Regime-conditional statistics confirm that the model captures genuine structural volatility heterogeneity: the Bear regime exhibits annualized realized volatility approximately 2.3× the Bull baseline, with negative mean returns. Across $k \in \{2, 3, 4, 5\}$, the raw BIC decreases monotonically; $k^* = 3$ is selected on economic grounds, as three regimes map directly to the canonical Bull/Neutral/Bear taxonomy without fragmenting regime structure—a choice the penalized criterion does not contradict.

The comparative analysis demonstrates that DTW hierarchical clustering, operating on 21-day rolling windows, improves materially over static K-Means by introducing temporal smoothing. However, the Gaussian HMM remains the superior framework for this application, offering explicit probabilistic persistence via the transition matrix, well-calibrated posterior regime probabilities for uncertainty quantification, and a principled real-time filtering procedure via the forward algorithm.

These findings validate the HMM pipeline as a suitable conditioning layer for the broader intraday momentum strategy, with clear operational protocols: full sizing in Bull regimes, reduced exposure in Neutral regimes, and strategy gating in Bear regimes.

A. REPOSITORY STRUCTURE AND EXECUTION

A.1. Key Dependencies

Table 7. Core Python Dependencies

Package	Purpose
`hmmlearn`	Gaussian HMM estimation (Baum–Welch, Viterbi)
`dtaidistance`	Dynamic Time Warping distance computation
`sklearn`	PCA, preprocessing, silhouette scoring
`pandas-datareader`	FRED API data retrieval
`yfinance`	Yahoo Finance market data retrieval
`statsmodels`	ADF and KPSS stationarity tests

A.2. Code Source

This project was developed entirely by the authors using the **Python** programming language and its associated scientific ecosystem. The complete source code, documentation, and implementation details are publicly available at the following repository:

https://github.com/blackswan-quants/marketregime_hmm

ABOUT THE AUTHORS

This project was developed jointly by **BlackSwan Quants** (BSQ), a quantitative finance student organization focused on quantitative research and algorithmic trading, and **PoliMi DataScientists** (PMDS), the data science student club of Politecnico di Milano. The two teams collaborated across model design, implementation, and validation.
BSQ: https://www.linkedin.com/company/blackswan-quants/
PMDS: https://www.linkedin.com/company/polimi-data-scientists/

REFERENCES

[1] A. Viterbi, "Error bounds for convolutional codes and an asymptotically optimum decoding algorithm", *IEEE Transactions on Information Theory*, vol. 13, no. 2, pp. 260–269, 1967.

[2] H. Sakoe and S. Chiba, "Dynamic programming algorithm optimization for spoken word recognition", *IEEE Transactions on Acoustics, Speech, and Signal Processing*, vol. 26, no. 1, pp. 43–49, 1978.

[3] R. F. Engle, "Autoregressive conditional heteroscedasticity with estimates of the variance of united kingdom inflation", *Econometrica*, vol. 50, no. 4, pp. 987–1007, 1982.

[4] C. R. Harvey, "The real term structure and consumption growth", *Journal of Financial Economics*, vol. 22, no. 2, pp. 305–333, 1988.

[5] L. R. Rabiner, "A tutorial on hidden markov models and selected applications in speech recognition", *Proceedings of the IEEE*, vol. 77, no. 2, pp. 257–286, 1989.

[6] N. Jegadeesh and S. Titman, "Returns to buying winners and selling losers: Implications for stock market efficiency", *The Journal of Finance*, vol. 48, no. 1, pp. 65–91, 1993.

[7] F. Petitjean, A. Ketterlin, and P. Gançarski, "A global averaging method for dynamic time warping, with applications to clustering", *Pattern Recognition*, vol. 44, no. 3, pp. 678–693, 2011.

[8] H. Holzmann and M. Schwaiger, "Identifying volatility regimes using hidden markov models", *Journal of Financial Econometrics*, vol. 14, no. 3, pp. 456–478, 2016.

[9] Y. Liu and Q. Wang, "Three-state hidden markov models for chinese stock index returns", *Emerging Markets Finance and Trade*, vol. 53, no. 5, pp. 1–20, 2017.

[10] J. Wang and K. Liu, "Hidden markov models for volatility regime detection in equity markets", *Applied Economics Letters*, vol. 24, no. 3, pp. 139–148, 2017.

[11] L. Chen, Y. Zhang, and H. Wang, "Regime switching in chinese stock markets: An hmm approach", *Quantitative Finance*, vol. 20, no. 7, pp. 827–863, 2020.

[12] H. Wang, L. Chen, and X. Zhao, "Regime-switching portfolio optimization with hmm-inferred states", *Journal of Asset Management*, vol. 21, no. 6, pp. 674–683, 2020.

[13] L. S. E. G. (LSEG), "Comparative analysis of hmm and clustering techniques on equity futures", LSEG Research, Tech. Rep., 2023.

[14] QuantConnect, *Intraday regime detection with hmm and technical indicators*, 2023. [Online]. Available: https://www.quantconnect.com/research.

[15] R. Ansari, "Volatility regime detection: Gaussian hmms versus wasserstein clustering", *SSRN*, 2025.

[16] F. Duriez, "Gaussian hmms for equity and cryptocurrency markets: Rolling vs full-sample fits", SSRN, Tech. Rep., 2025.